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The WNET Group

How to Make Your Digital Media Accessible for All Audiences

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1 in 4

American adults
has a disability

Source:
CDC, 2020
https://bit.ly/3BZBUfP

What is a screen reader?

"An assistive technology primarily used by people with vision impairments. It converts text, buttons, images and other screen elements into speech or braille."

Source:
AxessLab.com, November 15, 2019
Daniel Göransson

What we'll cover

- Graphic Design
- Web Design
- Digital & Social Video
- Social Media Posting

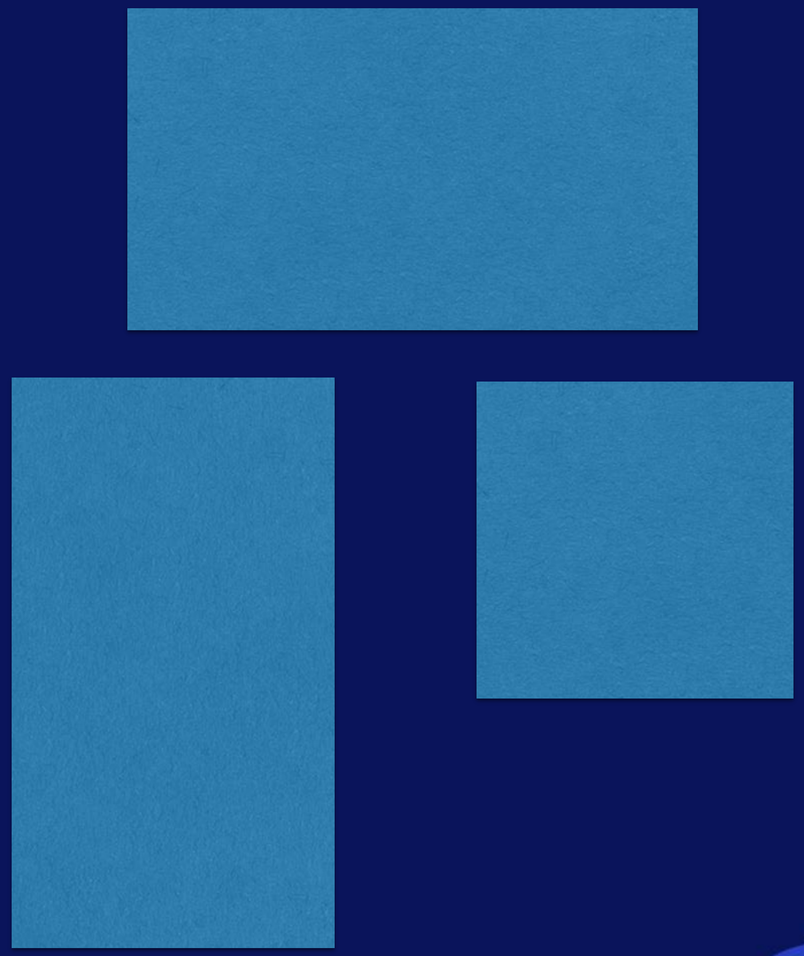
Designing for Accessibility

Content is everything

The WNET Group and the PBS family create a lot of awesome content. More and more of that content is being accessed beyond broadcast through station and program websites, OTT devices and a growing number of other platforms.

Know your platform

- Create images with the specific platform in mind.
- There is no one-size-fits-all image size for all social media platforms.
- Unexpected cropping can render your social graphics unreadable. This is a legibility issue for ALL users.



Designing compositions

- Instagram uses 1:1 ratio for static images. Instagram Stories use 9:16 ratio for static images.
- Facebook images can vary but square size (1:1) is a safe bet.
- Twitter uses horizontal 16:9 ratio for static images.
- When creating video content, leave room for video controls and space for an ASL interpreter.

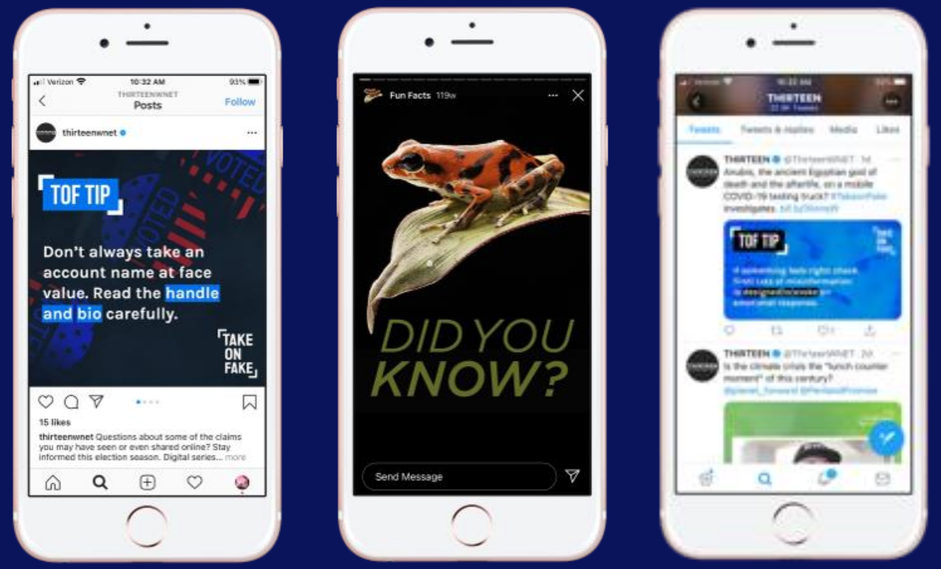


Image description: Three cell phones. On the first phone, an Instagram post is square, filling about half of the phone screen with the text "Take on Fake Tip: Don't always take an account name at face value. Read the handle and bio carefully." On the second phone, an Instagram Stories post is vertical, filling the entire phone screen with a photo of a frog and the text "Did you know?" On the third phone, a 16:9 graphic is shown on Twitter.

Optimal/not optimal fonts

SAN SERIF

Arial
Helvetica
Verdana
Tahoma

SERIF

Times New Roman
Georgia

SLAB SERIF

Arvo
Museo Slab

DECORATIVE FONTS

Way too Fancy

HARD tO READ

SERIOUSLY?

Image description: Samples of various font styles including san serif, serif, slab serif and hard to read fonts: a brush-stroke script, a newspaper-cutout style and a gameshow style that is nearly unreadable.

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Helpful ways to increase legibility

- Kerning between letters is important, particularly between *R*s and *N*s.
- A word like *modern* can be read as *modem* if the letters are too tightly kerned.

FAILS

Modem

PASSES

Modern

Helpful ways to increase legibility

- Line height (or line spacing) should not be tightly stacked.
- Line height should be at least 1.5 times the font size.
- Spacing following paragraphs should be at least 2 times the font size.

FAILS

Even an accessible typeface like Arial can become harder to read when the spacing between the lines is too tight.

PASSES

Even an accessible typeface like Arial can become harder to read when the spacing between the lines is too tight.

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Helpful ways to increase legibility

- Left align your text. It is easier to read as each line
  of text starts near the same position as the previous line.
- People with cognitive disabilities may have trouble with blocks of text that are fully justified on both the left and right side.

Left aligned text is easier to read than centered text for paragraphs. This is because when you center your text, the starting place of each line changes. This forces your users to work harder to find where each line begins to continue reading. Without a straight left edge, there is no consistent place where users can move their eyes to when they complete each line.

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Helpful ways to increase legibility

- Don’t make lines of text too long or too short. The number of characters per line of text should not exceed 80. The generally acceptable range is between 45-75 characters.

- Shorter pieces of text are fine for captions and forms.

FAILS

Typographers have been writing about the relationship between horizontal and vertical eye motion for almost a century. If a casual reader gets tired of reading a long horizontal line, then they’re more likely to skim the left edge of the text. If an engaged reader gets tired of reading a long horizontal line, then they’re more likely to accidentally read the same line of text twice (a phenomenon known as “doubling”).

PASSES

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Design tips for legibility

- Text on a photo can be difficult to read.
- Darken or lighten the area behind text to increase legibility of overlaid text.
- Drop shadows help.

Image description: An image of Helen Keller sitting and smelling a flower in a light pink dress, overlaid with text about an upcoming episode. The words in white against her clothing are difficult to read. When a black gradient is added to the bottom of the image behind the text, the words are legible.

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Not everybody sees **color** the same way.

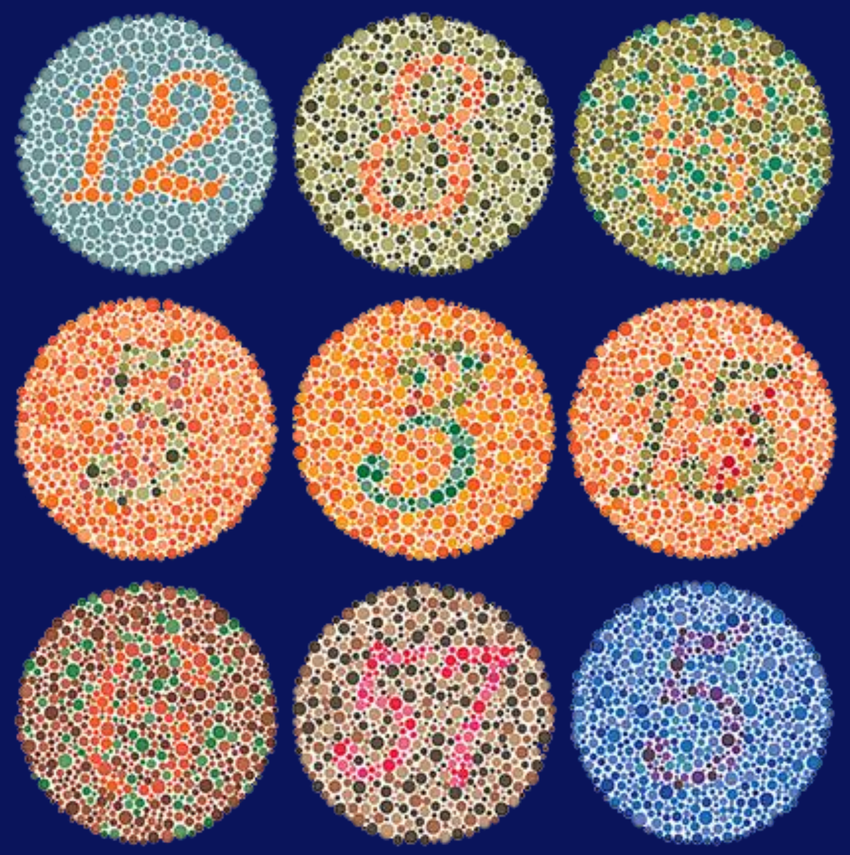
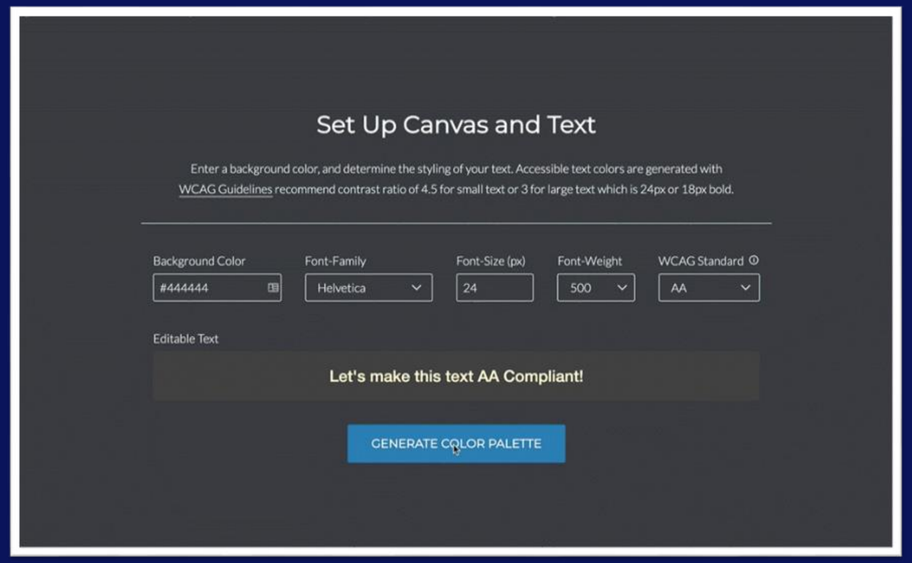


Image description: A series of circles in a color deficiency test shows that some combinations of colors are very difficult to distinguish.

Creating color palettes

For WCAG Level AA accessibility compliance, text and images of text must meet a contrast ratio of at least 4:5:1. Large text and images of large-scale text must be at 3:1. But don’t panic! Color contrast can easily be tested for accessibility with various online tools.



Did you know that logos ***do not need to conform*** to WCAG color contrast requirements?

That being said, ***we should take it upon ourselves*** to create logos and title treatments that are as accessible as possible to our audience.

Website Design for Accessibility

Use of color and indicators

Color should not be used as the ***only*** visual means of conveying information, indicating an action, prompting a response or distinguishing a visual element.

DEFAULT STATE

Email

FAILS

Email

PASSES

Email required

Use of color and indicators

Assigning proper heading-level text structures your content for users.

<h1> Main Heading </h1>

<h2> Sub Heading </h2>

<h3> Sub Sub Heading </h3>

<h2> Sub Heading </h2>

<h3> Sub Sub Heading </h3>

<h4> Sub Sub Sub Heading </h4>

Alternative text for images

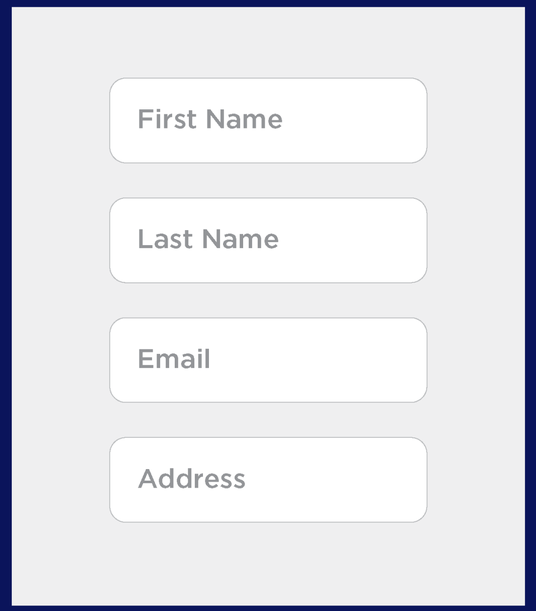
- Include alt text for any image you use on your website and social media to aid users who are blind or with low vision, screen readers or even slow internet connections.
- Purely decorative images don’t need alt text.
- Keep alt text concise.

Image description: Helen Keller reading a braille book while petting a dog.

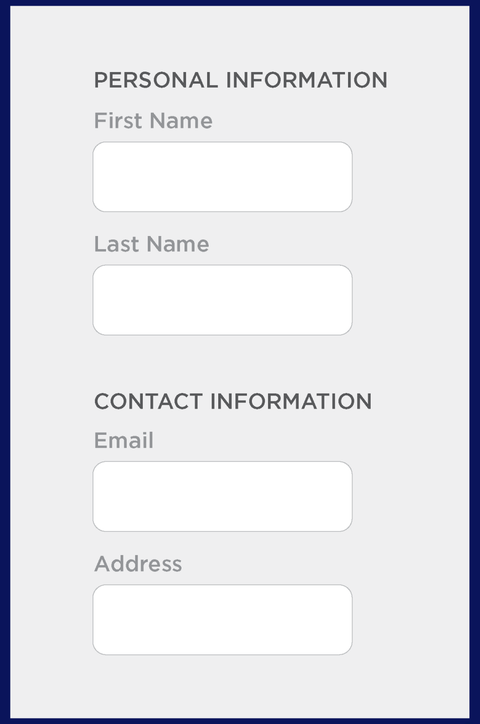
Labeling forms and inputs

Give clear instructions on how to fill out forms visually by using label text versus placeholder text that disappears once something is typed.

FAILS

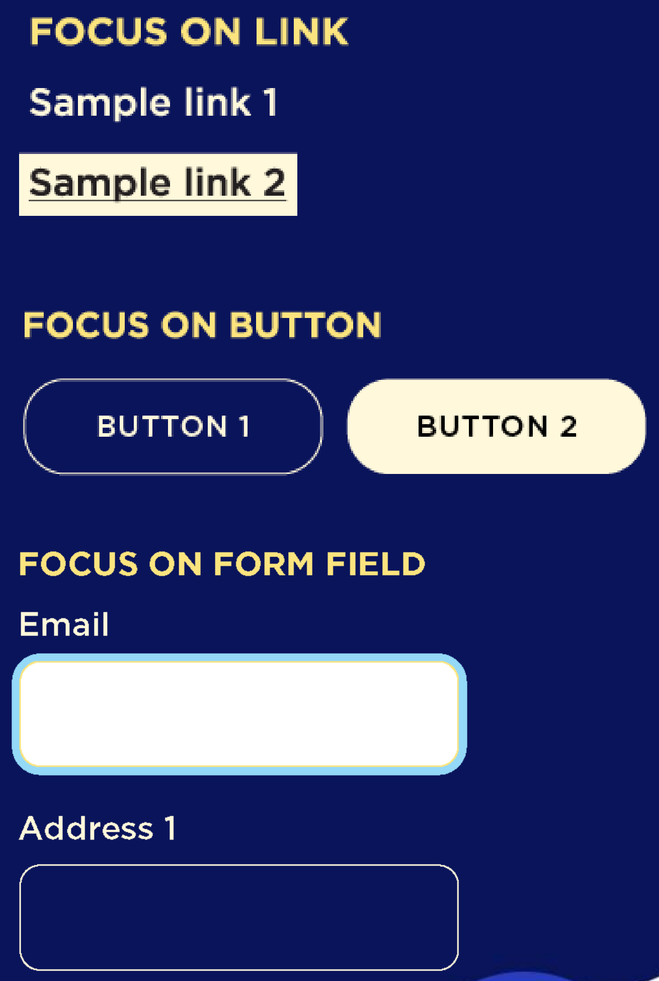


PASSES



Usable focus states and keyboard navigation

Focus indicators are a great way to help users know which element has the keyboard focus and understand where they are when navigating your site.



Digital & Social Video Accessibility

Produce with accessibility in mind

- Representation matters both in front of and behind the camera.
- Be sure to consider the accessibility needs of all involved from pre-production through post-production.

Image description: A photograph of a woman being filmed. She is sitting in a library, surrounded by bookshelves and is smiling at the camera.

Don't create a TASTY-style video

Don't rely solely on a combination of visuals and music, as that makes your content **inaccessible** to someone who is blind or with low vision.

Image description: An image of hands scooping an avocado into a glass bowl on a marble counter. In the top-right corner is a tomato, cilantro, avocado and a lime. In the top left is green text that says "avocados," and underneath is the number 3.

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Incorporate audio description

"Descriptive audio is the additional audio content that **describes aspects of your video that are purely visual** and not accessible to blind or [with low vision]."

-UC Berkeley

Image description: A man sits at his computer with a microphone facing towards him. He has two monitors. On one screen, a cue sheet; on the other, a video file among other windows.

Create your own audio description

- Add audio description to your videos to reach a blind and low-vision audience.
- When filming, have subjects identify themselves and their surroundings.
- Capture as much good, clean location sound as possible.
- Write short, concise sentences.
- Leave room in the edit.
- Preserve the same discoveries, emotional beats, pauses and tension for the audience.

Image description: Square screengrab of a social-optimized trailer or "Becoming Helen Keller," featuring a black and white image of Helen Keller, open captions and an on-screen ASL interpreter.

Or add audio description later

- Cast audio description narrators that match the content.
- When a description track is not possible, share a second version of your video with audio description baked in.
- Write simply, clearly and concisely so that the description is easy to comprehend.
- Some description may be read briskly to fit it into the content.

Create extended audio description

- Extended audio description **temporarily pauses the audio and video to** allow critical information to be delivered.
- Create a second version of your video with extra description.
- This can make your content more accessible for people who need to hear not only the dialogue but also the context and other aspects of the video.

Create open captions

- Add **open captions** to your videos to reach a deaf and hard-of-hearing audience.
- Include dialogue, music and sound effects cues.
- Open captions will also benefit the 85% of Facebook users who watch videos without sound.

Image description: Square screengrab of a social-optimized clip from "Becoming Helen Keller," featuring open captions and Rebecca Alexander and an ASL interpreter on-screen.

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Record on-screen ASL interpretation

- Why not just use closed/open captions?
  - ➢ ASL has its own grammar and syntax.
  - ➢ It is seen, not read.
- Many prefer Deaf interpreters.
- Leave space on-screen for ASL.
- Make your edit points accurate.

Image description: Screengrab from *Renegades – Kitty O'Neil*, featuring speed racer Kitty O'Neil, with an ASL interpreter on-screen in the bottom right corner.

Avoid flashing lights and quick cuts

- Avoid using flashing content or overly fast cuts in videos and GIFs.
- Ensure your edits meet the "three flashes or below" threshold.

Do not use videos that have more than three flashes within a period of one second, as this can provoke seizures in some users with seizure disorders.

-Bureau of Internet Accessibility

Build combined transcripts

- Publish combined transcripts alongside your video content to reach a DeafBlind audience.
- This is a transcript with the text version of the speech and non-speech audio information.
- Includes **all** the audio and visual information a user would get from watching and hearing the video.

Social Media Posting

Use CamelCase hashtags

- Capitalize the first word of any multi-word hashtag, rather than keep all letters in lower case.

#HelenKeller instead of #helenkeller

- Allows for easier reading.

Write image descriptions in your captions

- Describe the image so key information can be communicated to those who are blind or with low vision.
- Indicating “PIC” or “VID” helps to identify the content type.

Image description: An *American Masters* Instagram post that includes the following image description in the caption: "PIC 1: Kitty O'Neil in a Racecar Made by Ky Michaelson."

Link to post: https://bit.ly/3vejDsu

Include a video transcript in your caption

- Add written/overlaid text and dialogue to the transcript; distinguish when the text is voiceover and when it's on-screen titles.
- Note that some people may find large blocks of text difficult; unfortunately, there is no common best practice.
- Note: Some platforms' character limits may not allow for this, and some platforms may reduce your reach with long copy; test and learn.

Image description: An *American Masters* Facebook post featuring a video clip from a documentary about Helen Keller. The accompanying caption includes the full transcript of the video.

Link to post: https://bit.ly/30lvUji

Keep all tags at the end of the caption

Placing all at-tags/handles and hashtags at the end of the caption will allow screen readers to read the copy in a smooth, uninterrupted flow.

I learned about #helenkeller while watching @PBSAmerMasters on @PBS.

Should be…

I learned about Helen Keller while watching American Masters on PBS. #HelenKeller @PBSAmerMasters @PBS

Keep it short and avoid all-caps

- Captions should be short and to the point.
- Longer sentences can be difficult to follow.
- Shorten long links with Bitly or another link shortener program.
- Acronyms like BTW (by the way) or LOL (laugh out loud) may confuse screen readers.
- Sentence case makes words easier to read, especially for some individuals with learning disabilities.

Repeat overlaid text in your captions

- Overlaid text will not be identified by screen readers.
- If an image has text on it, repeat the same text in the post caption.

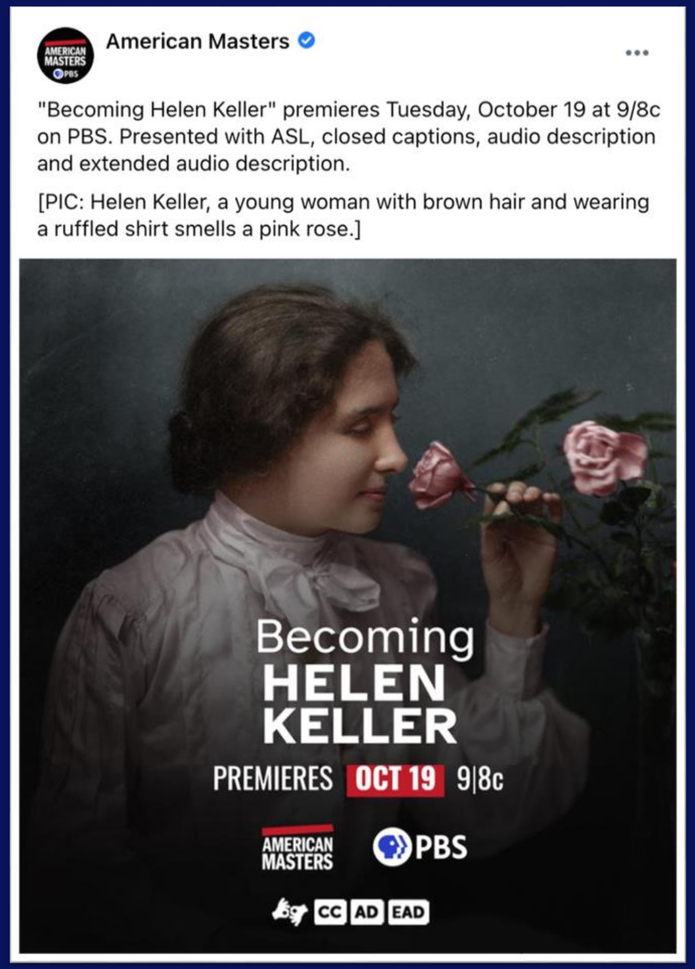

Image description: An *American Masters* Facebook post featuring a tune-in graphic with overlaid copy and a caption that repeats the information on the graphic. It reads, "'Becoming Helen Keller' premieres Tuesday, October 19 at 9/8c on PBS. Presented with ASL, closed captions, audio description and extended audio description. [PIC: Helen Keller, a young woman with brown hair and wearing a ruffled shirt smells a pink rose.]

Design tip

Test your design on a phone… not everybody is seeing it on a gigantic screen.

Image description: A big computer monitor shows the text "Looks great here!" while a cell phone screen shows similar text as a question, "Does it still look great here?"

And one more thing…

- “Nothing about us, without us.”
- Try to work with consultants, advisers and creative collaborators from within the disability community.
- Build in more review time on assets so you can make versions and cuts accessible.
- Provide timed transcripts for cuts being reviewed.

Thanks!

Appendix

WCAG website

- The Web Content Accessibility Guidelines (WCAG) is defined by the World Wide Web Consortium (W3C).

Color contrast tools

Tools built for designers and developers to test contrast between two colors for compliance within the WCAG.

- https://contrastchecker.com/
- https://colorshark.io/
- https://webaim.org/resources/contrastchecker/

Color contrast tools

- http://colorsafe.co/
  A tool built for designers and developers to create more complex color palettes compliant with the WCAG.

- https://www.color-blindness.com/coblis-color-blindness-simulator/
  This tool is built for designers to test how an image will appear to users with eight different types of color blindness. This is especially helpful if your image has text overlays.

List of Video Resources

W3C Web Accessibility Initiative: Making Audio and Video Media Accessible

Bureau of Internet Accessibility: Checklist for Creating Accessible Videos

American Council for the Blind: All About Audio Description

Hootsuite: Everything You Need to Know About Inclusive Design for Social Media

Frame.io: Design for the Edges: Video for the Blind and Deaf

3Play Media: Social Video: Why Aren't We Making It Accessible to Blind People?

Consultants for audio description, ASL interpretation, and other services

- https://bridgemultimedia.com/ (Audio description, ASL, transcription, consultation on best practices)
- https://d-pan.org/ (Deaf Professional Arts Network, specializing in ASL interpretation videos)

Social Media Posting

Avoid using emoticons

- Emoticons that require typing various characters to create an image can be confusing.
- Emojis are okay to use because screen readers will read their matching text description (for example, the emoji on the right will read as “shrug”).

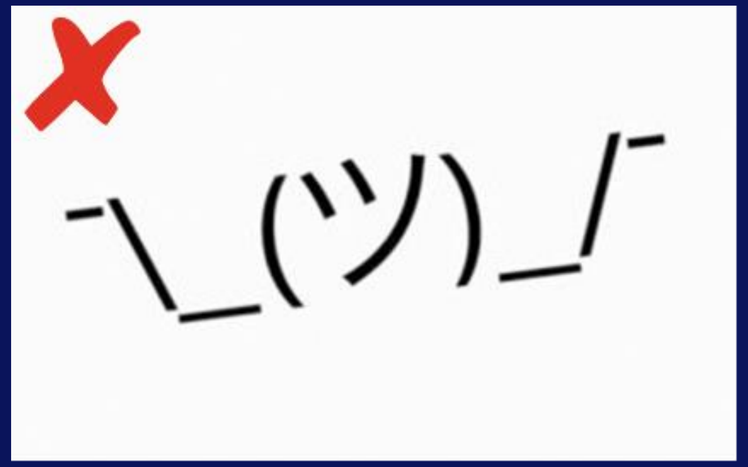

Image description: Two images. The image on the left is of the “shruggie” emoticon, which includes several typed symbols that come together to form the likeness of a person’s head and arms, with arms shrugged. On the top-left corner is a red X.
The image on the right is the “shrug” emoji, which is a more detailed illustration of a woman making the same gesture with her arms. On the top-left corner is a green checkmark.

Social Media Posting

Add alt text to images

- Screen readers will read your brief description of the image for people who are blind or with low vision.
- Most social media platforms, including Facebook, Twitter, Instagram and LinkedIn offer this option, some with character limits.

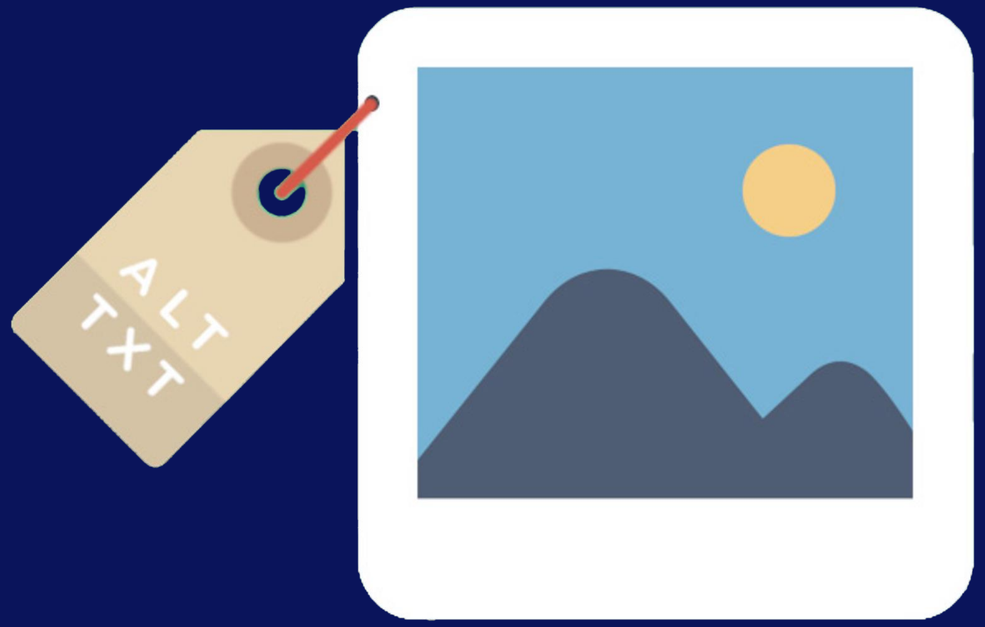

Image description: A simple illustration of a photo of mountains and the sun. A tag hangs off the photo. On it is written, “ALT TXT.”

Resource:
Learn how to add alt text on Facebook, Instagram, Twitter and LinkedIn: https://bit.ly/36IhVVO

Social Media Posting

Include a website link and means of contacting your organization

Adding these elements to your social media page's "about" section or bio will benefit individuals who need to reach the organization.

Image description: The *American Masters* Instagram bio, featuring the show's logo, an e-mail address and phone number.

Resources for hosting virtual and in-person events

- Planning Accessible Meetings and Events: A Toolkit (Commission on Disability Rights)
- Planning Accessible Meetings and Conferences: A Suggested Checklist and Guide (University of Maine)
- Accessibility Checklist for University Events (University of Connecticut)
- Accessible Meeting and Event Checklist (Cornell University)
- Hosting accessible online meetings (University of Washington)
- How to Make Your Virtual Meetings Accessible to All (Idealist)
- Make Your Events Truly Accessible and Inclusive (The Chicago Community Trust)